Introduction to Computer Science
Fall 2026
Homework 1 Answer

Problem 1: Is This Shuffle Random?

No. For any $n \geq 3$, the algorithm has n^n equally likely sequences of random choices. If all $n!$ permutations were equally likely, then $n^n/n!$ is an integer, thus $n-1|n!|n^n$, hence $n-1|1^n$, contradiction. Therefore, the algorithm does not generate all permutations with equal probability for any $n \geq 3$.

Problem 2: Transformer

(a)

Let $u_i = v_i/\|v_i\|$ and $u_j = v_j/\|v_j\|$. By symmetry, each coordinate of u_i has mean zero, so

$$\mathbb{E}[\cos(v_i, v_j)] = \mathbb{E}[u_i^\top u_j] = \mathbb{E}[u_i^\top (-u_j)] = -\mathbb{E}[u_i^\top u_j],$$

thus

$$\mathbb{E}[\cos(v_i, v_j)] = 0.$$

Since $\sum_{r=1}^{d} u_{i,r}^2 = 1$ and all coordinates are identically distributed, $\mathbb{E}[u_{i,r}^2] = 1/d$. Also, symmetry gives $\mathbb{E}[u_{i,r}u_{i,s}] = 0$ for $r \neq s$. Using independence, we obtain

$$\begin{aligned}
\mathrm{Var}(\cos(v_i, v_j)) &= \mathbb{E}\left[\left(\sum_{r=1}^{d} u_{i,r}u_{j,r}\right)^2\right] \\
&= \sum_{r=1}^{d} \mathbb{E}[u_{i,r}^2]\mathbb{E}[u_{j,r}^2] \\
&= d\left(\frac{1}{d}\right)^2 = \frac{1}{768}.
\end{aligned}$$

(b)

Computing q_i, k_i, and v_i for all n tokens involves $3n$ matrix-vector products, each requiring d^2 multiplications and $d(d-1)$ additions. This contributes $3n(2d^2 - d)$ operations.

Evaluating the n^2 attention scores requires $n^2(2d-1)$ operations. For each output y_i, the weighted sum requires nd multiplications and $(n-1)d$ additions, giving $nd(2n-1)$ operations across all outputs.

Thus,

$$\begin{aligned}
T(n, d) &= 3n(2d^2 - d) + n^2(2d-1) + nd(2n-1) \\
&= 6nd^2 + 4n^2d - 4nd - n^2.
\end{aligned}$$

(c)

For $n = 10^5$ and $d = 10^4$, part (b) gives $T(n, d) \approx 4.6 \times 10^{14}$ operations per layer. Therefore, the daily energy consumption of 100 layers serving 7×10^9 queries is

$$\begin{aligned} E &= 100 \cdot (7 \times 10^9) \cdot (4.6 \times 10^{14}) \cdot (2 \times 10^{-10}) \\ &\approx 6.44 \times 10^{16} \text{ J}. \end{aligned}$$

Problem 3: Quantum Key Distribution

(a)

1. When Alice and Bob choose the same basis, the transmitted qubit is an eigenstate of Bob's measurement basis. Thus, Bob obtains Alice's bit with probability 1, and no errors occur in the sifted key.

2. Since both bases are chosen independently and uniformly,

$$P(\text{same basis}) = P(X, X) + P(Z, Z) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Hence, the expected length of the sifted key is $N/2$.

(b)

For each retained position, Eve chooses the correct basis with probability $1/2$, causing no error. Otherwise, her measurement destroys the original bit information, and Bob's result disagrees with Alice's bit with probability $1/2$.

Therefore, the expected error rate is

$$P(\text{error}) = P(\text{wrong basis})P(\text{error} \mid \text{wrong basis}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

(c)

From part (b), each compared bit reveals an error with probability $1/4$. Since errors are independent, the probability that none of the m sampled bits reveals Eve is $(3/4)^m$. Thus,

$$P(\text{detected}) = 1 - P(\text{no errors}) = 1 - \left(\frac{3}{4}\right)^m.$$

As $m \to \infty$, this probability approaches 1 exponentially fast.